Praktikum 18
Konfigurasi Pengalamatan IPv6

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MATA KULIAH	:	Praktikum Jaringan Komputer
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D4 TEKNIK KOMPUTER
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA
2025

Konfigurasi Pengalamatan IPv6

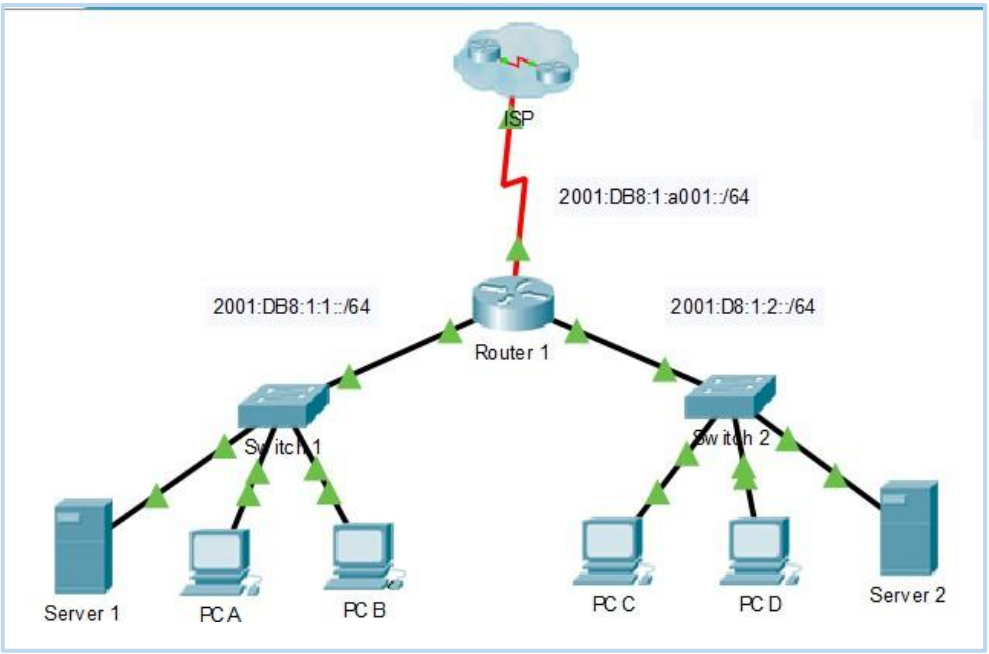
I. Tujuan Pembelajaran

1. Mahasiswa dapat menjelaskan konsep pengalamatan IPv6
2. Mahasiswa dapat melakukan konfigurasi pengalamatan IPv6 pada *Cisco Router*

II. Perangkat

1. Laptop
2. Packet Tracer

III. Topologi Jaringan

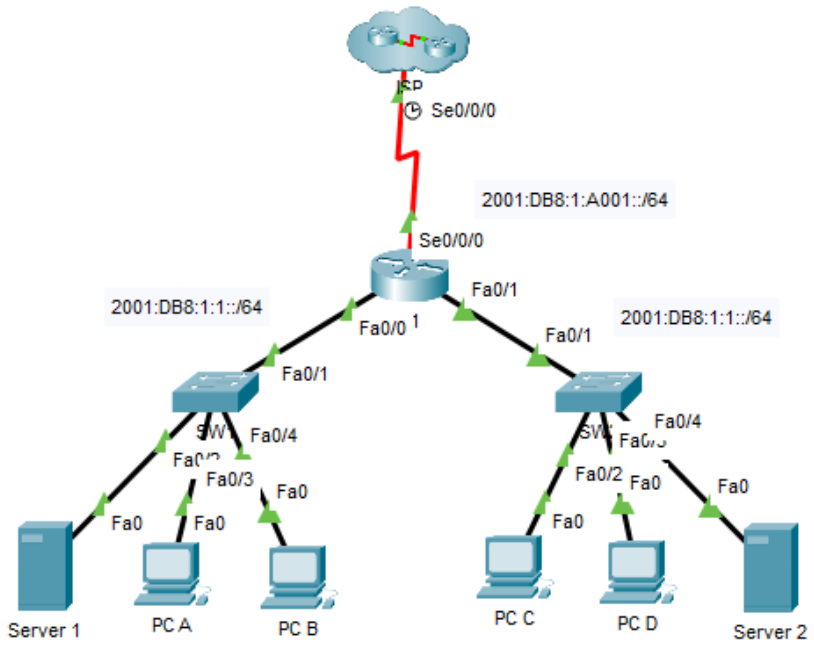


End Devices		
Nama Perangkat	IPv6 / *Prefix*	*Default Gateway*
Server 1	2001:DB8:1:1::4 / 64	FE80::1
PC A	2001:DB8:1:1::3 / 64	FE80::1
PC B	2001:DB8:1:1::2 / 64	FE80::1
PC C	2001:DB8:1:2::2 / 64	FE80::1
PC D	2001:DB8:1:2::3 / 64	FE80::1
Server 2	2001:DB8:1:2::4 / 64	FE80::1
Routers		
Nama Perangkat	IPv6 / *Prefix*	*Interface*
R1	2001:DB8:1:1::1/64	FastEthernet0/0
	2001:DB8:1:2::1/64	FastEthernet0/1
	2001:DB8:1:A001::2/64	Serial0/0/0
ISP	2001:DB8:1:A001::1/64	Serial0/0/0

IV. Prosedur Percobaan

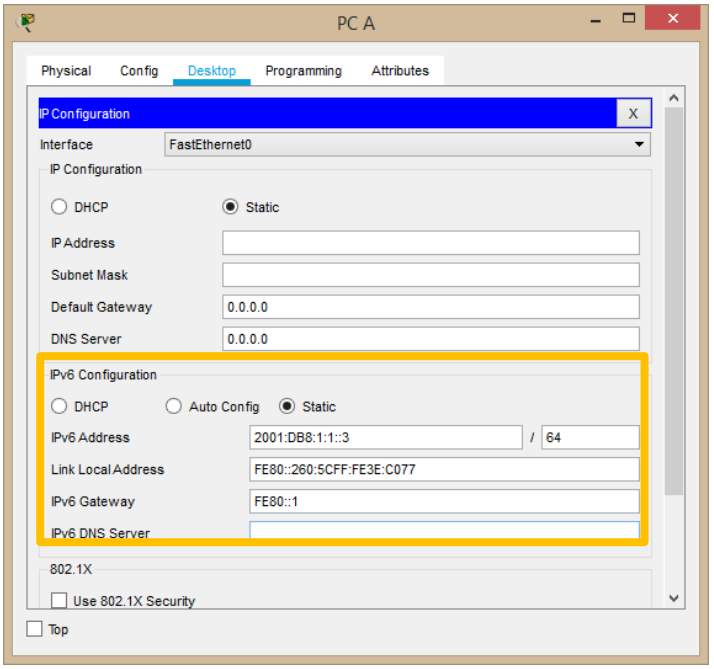
1. Buat topologi jaringan seperti yang ditunjukkan pada sub bab topologi jaringan di atas. Topologi dibangun pada simulator *Packet Tracer* dengan perangkat yang dibutuhkan yaitu :
   a. *End devices* : PC, *server*
   b. *Network devices* : *switch, router*
   c. *Connections* : *copper straight-through, serial*

   Hasil topologi:



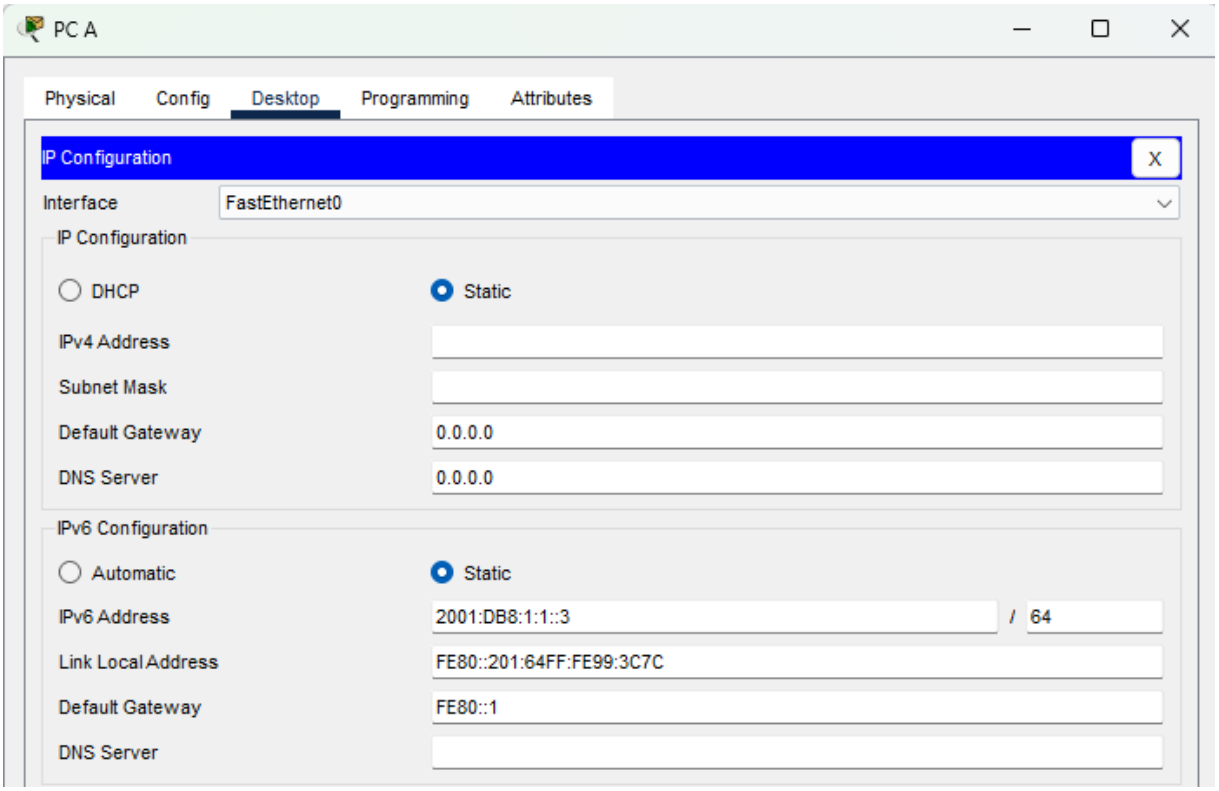
2. Lakukan Konfigurasi IPv6 pada semua *client*

   Contoh Konfigurasi IPv6 pada PC A:

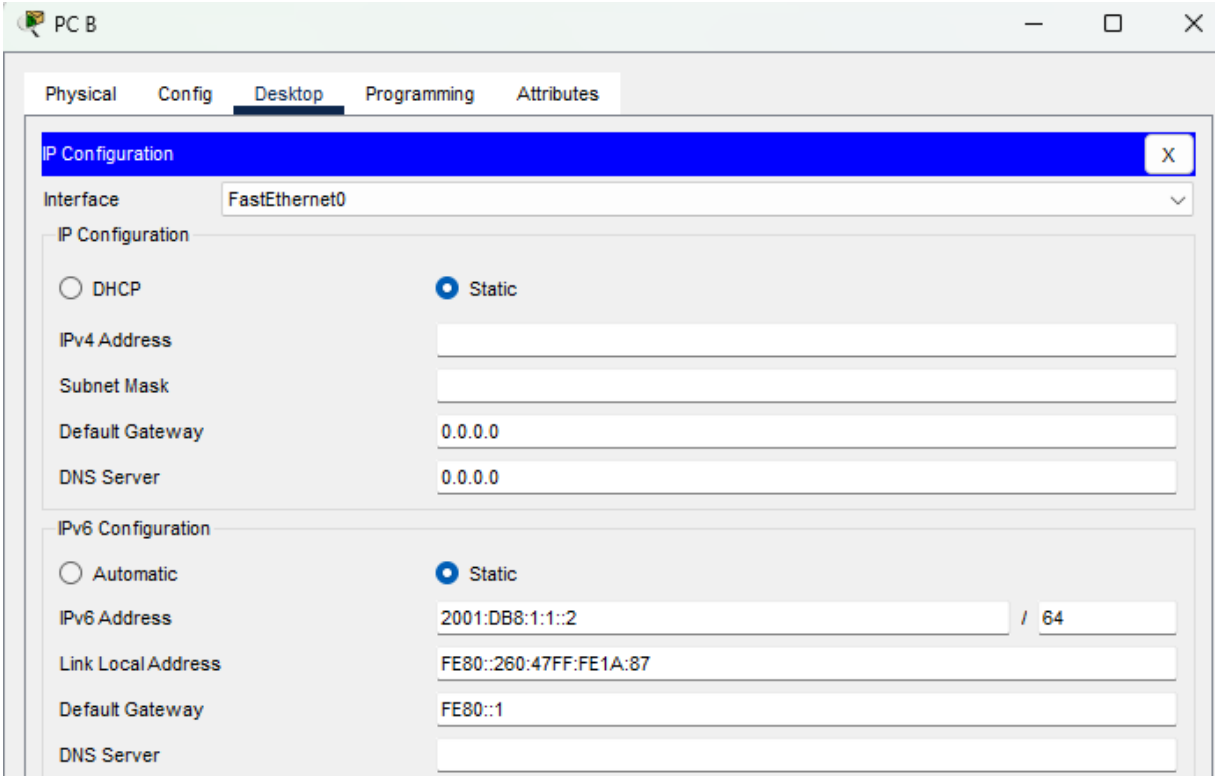


Hasil:

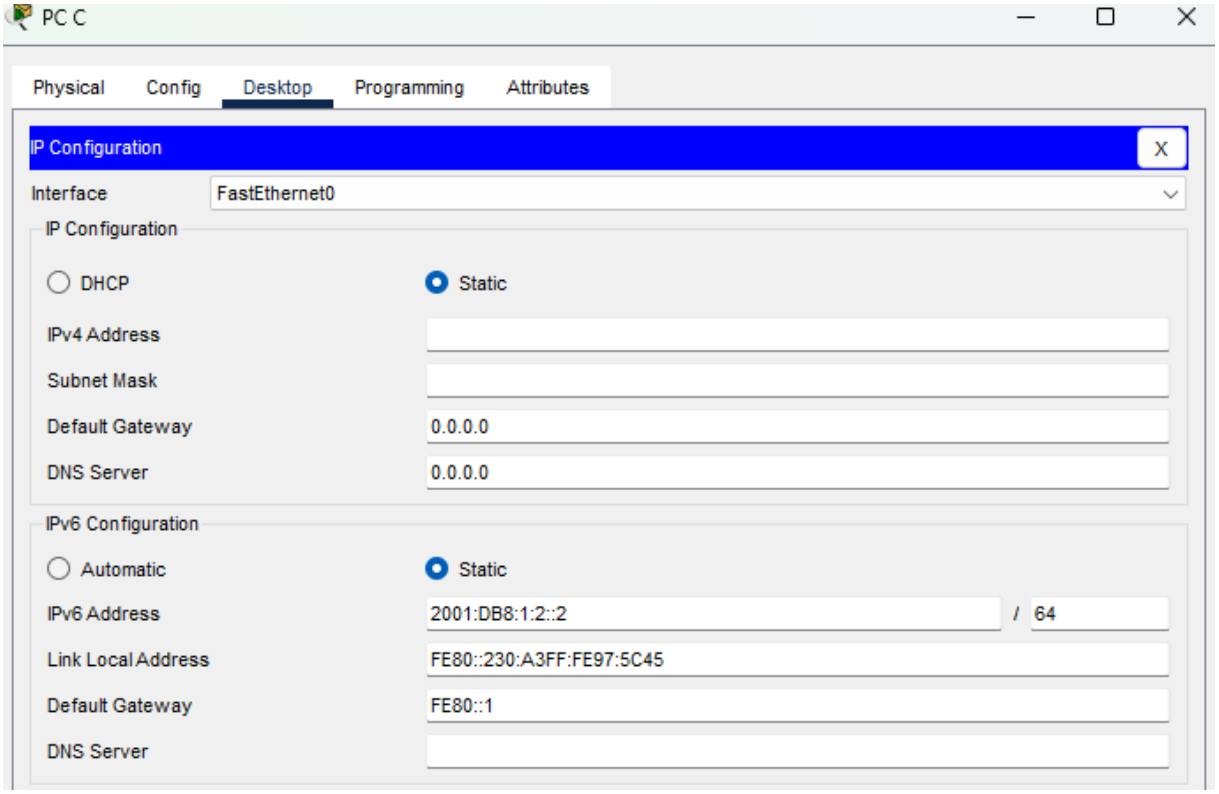
PC A



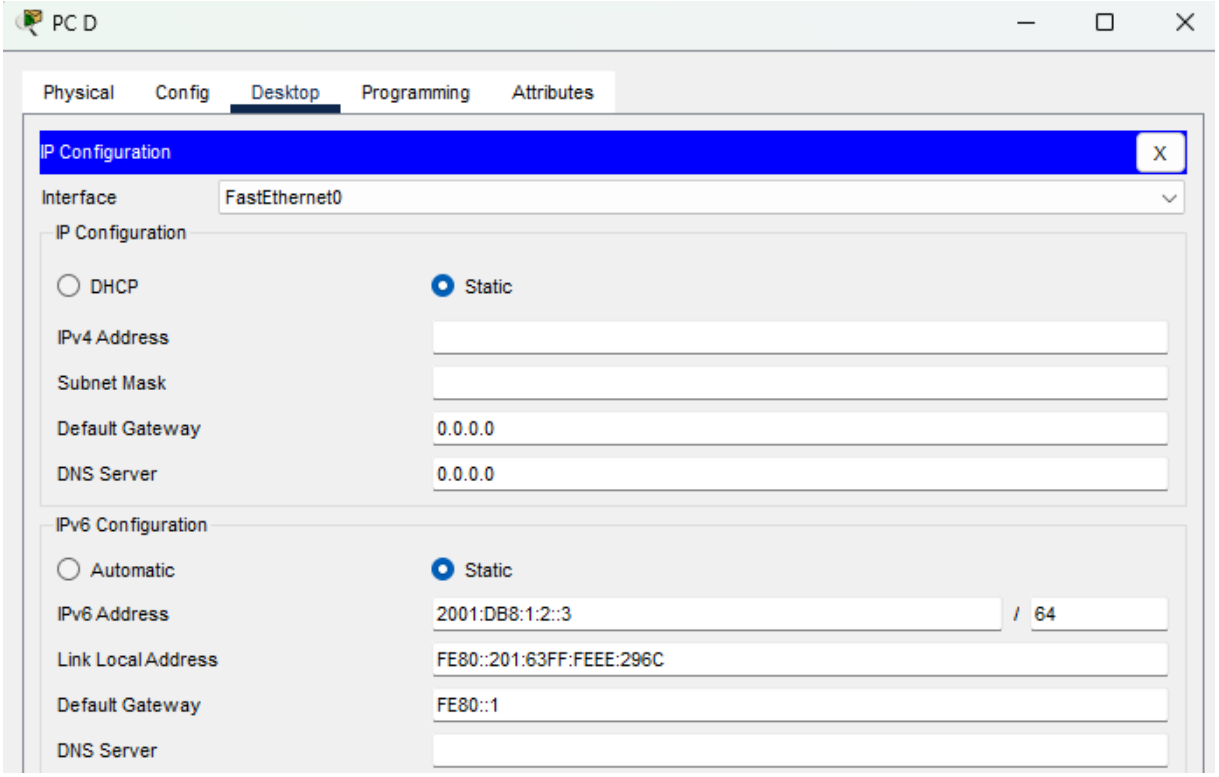
PC B



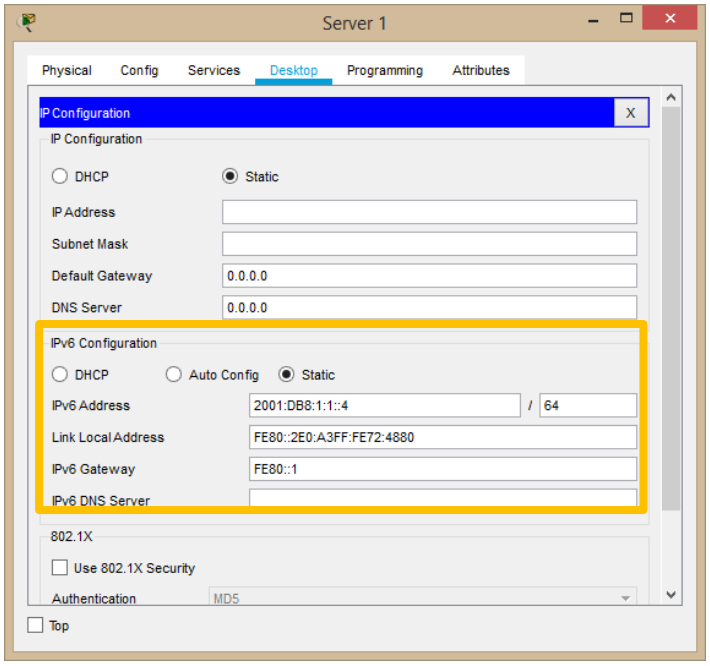
PC C

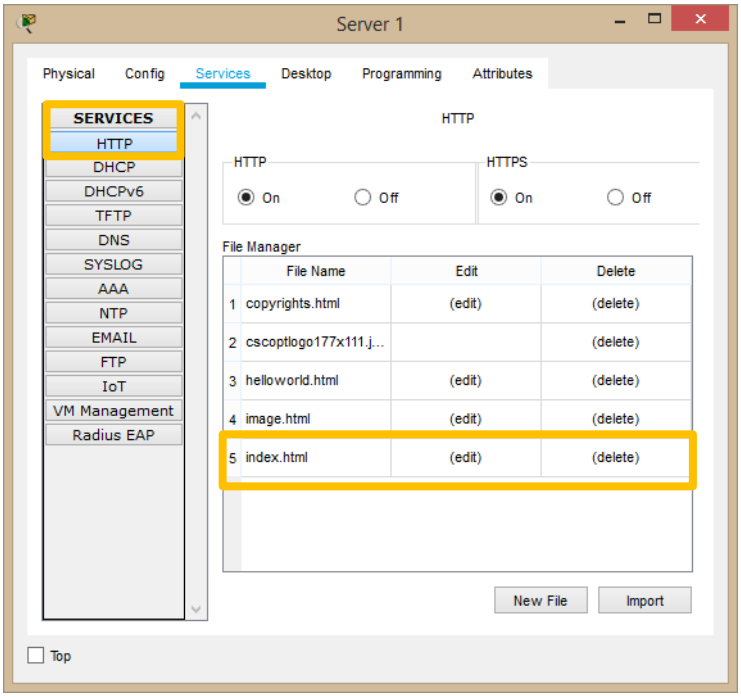

PC C
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1:2::2
/ 64
Link Local Address
FE80::230:A3FF:FE97:5C45
Default Gateway
FE80::1
DNS Server

PC D


PC D
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1:2::3
/ 64
Link Local Address
FE80::201:63FF:FEEE:296C
Default Gateway
FE80::1
DNS Server

3. Lakukan Konfigurasi IPv6 pada semua *server*

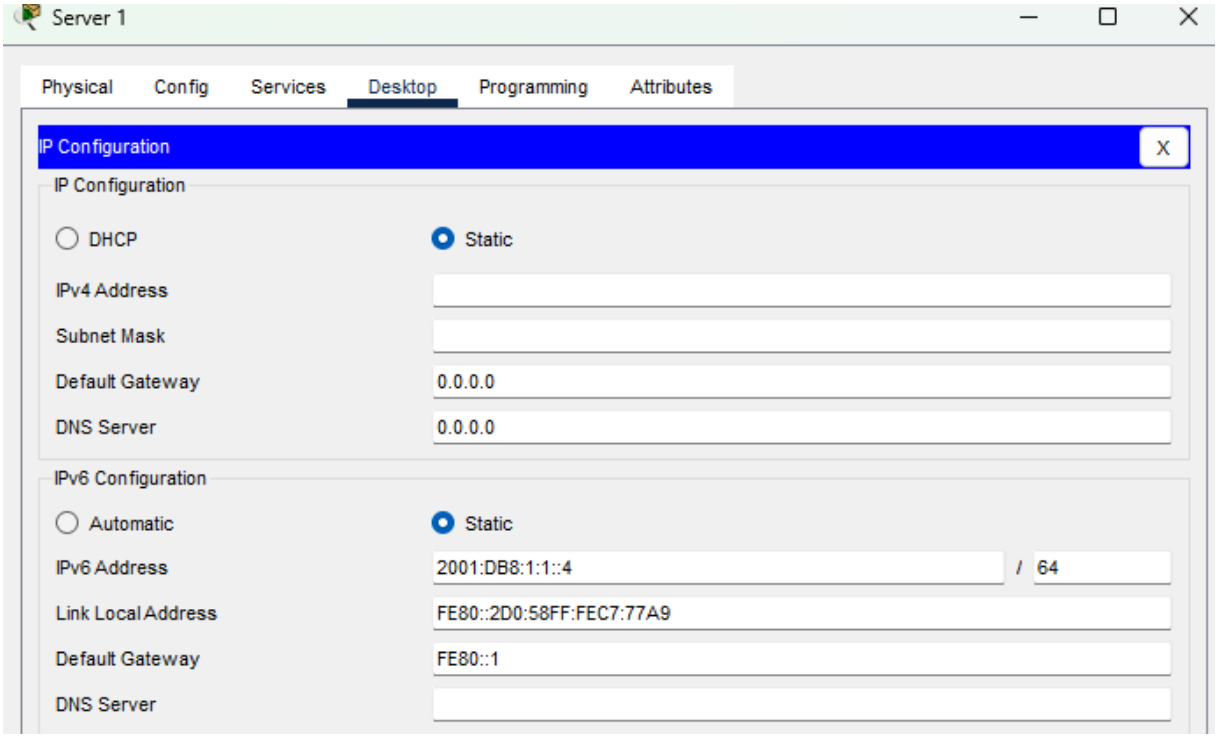




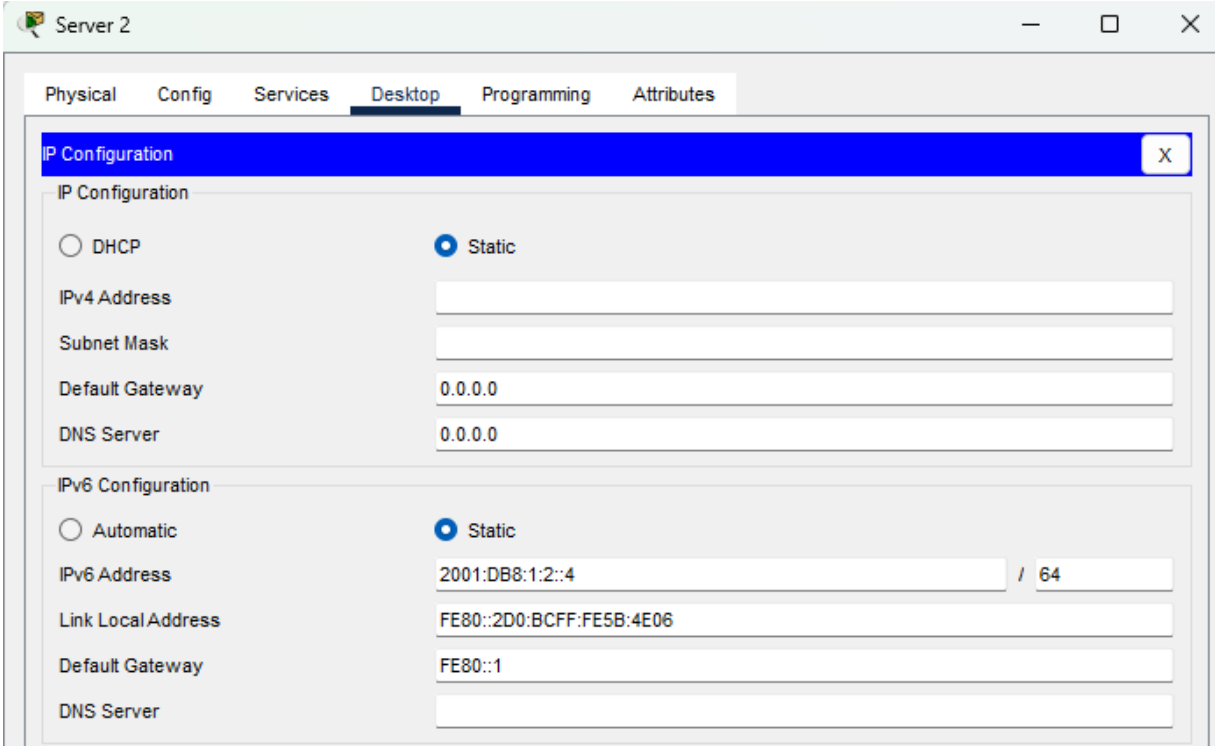
Contoh Konfigurasi pada Server 1:

Setelah melakukan konfigurasi pengalamatan IPv6, anda dapat melakukan konfigurasi untuk layanan yang akan disediakan oleh server. Pada percobaan ini akan menggunakan HTTP, dimana anda dapat mengedit file index.html ataupun menambah dan menghapus file pada *file manager*.

Server1



Server 2

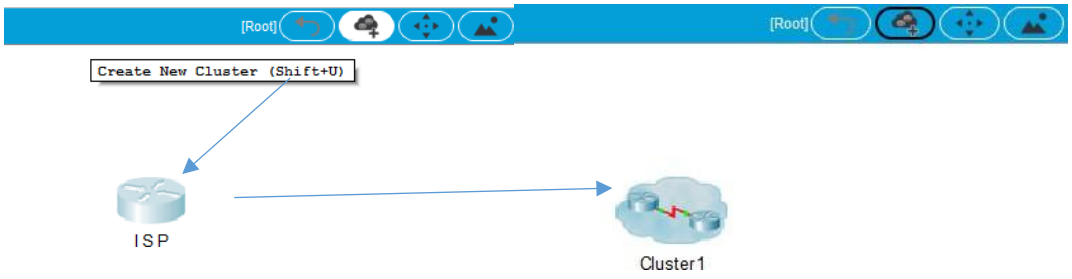


4. Lakukan Konfigurasi IPv6 pada semua *router*

   Konfigurasi pada Pengalamatan IPv6 pada *interface fastEthernet*0/0 di Router 1 :

```
Router>enable
Router#configure terminal
Router(config)# ipv6 unicast-routing
Router(config-if)#interface fastEthernet0/0
Router(config-if)#ipv6 enable
Router(config-if)#ipv6 address 2001:DB8:1:1::1/64
Router(config-if)#ipv6 address fe80::1 link-local
Router(config-if)#no shutdown
```

   Lakukan konfigurasi pengalamatan IPv6 untuk semua *interface* di Router 1. Kemudian lanjutkan konfigurasi pengalamatan IPv6 untuk ISP.



   Konfigurasi pada Pengalamatan IPv6 pada *interface* serial0/0/0 di ISP :

```
Router>enable
Router#configure terminal
Router(config)# ipv6 unicast-routing
Router(config-if)#interface serial0/0/0
Router(config-if)#ipv6 enable
Router(config-if)#ipv6 address 2001:DB8:1:A001::1/64
Router(config-if)#ipv6 address fe80::1 link-local
Router(config-if)#no shutdown
```

   Selanjutnya dapat dilakukan konfigurasi routing statis, agar ISP dapat terhubung ke semua *client*

```
Router#configure terminal
Router(config)#ipv6 unicast-routing
Router(config)#ipv6 route 2001:db8:1:1::/64 serial0/0/0 fe80::1
Router(config)#ipv6 route 2001:db8:1:2::/64 serial0/0/0 fe80::1
Router(config)#exit
```

Router 1:

```
R1(config)#ipv6 unicast-r
R1(config)#int f0/0
R1(config-if)#ipv6 en
R1(config-if)#ipv6 add 2001:DB8:1:1::1/64
R1(config-if)#ipv6 add FE80::1 link-l
R1(config-if)#no sh
R1(config-if)#int f0/1
R1(config-if)#ipv6 en
R1(config-if)#ipv6 add 2001:DB8:1:2::1/64
R1(config-if)#ipv6 add FE80::1 link-l
R1(config-if)#no sh
R1(config-if)#int s0/0/0
R1(config-if)#ipv6 en
R1(config-if)#ipv6 add 2001:DB8:1:A001::2/64
R1(config-if)#ipv6 add FE80::1 link-l
R1(config-if)#no sh
```

Router ISP:

```
ISP(config)#ipv6 unicast-r
ISP(config)#int s0/0/0
ISP(config-if)#ipv6 en
ISP(config-if)#ipv6 add 2001:DB8:1:A001::1/64
ISP(config-if)#ipv6 add fe80::1 link-l
ISP(config-if)#no sh
ISP(config-if)#ipv6 unicast-r
ISP(config)#ipv6 route 2001:db8:1:1::/64 s0/0/0 fe80::1
ISP(config)#ipv6 route 2001:db8:1:2::/64 s0/0/0 fe80::1
```

5. Lakukan pengecekan konfigurasi pada semua router untuk *interface* dan tabel *routing*

Pada Router 1

```
Router>enable
Router#show ipv6 interface brief
FastEthernet0/0 [up/up] FE80::1
      2001:DB8:1:1::1
FastEthernet0/1 [up/up] FE80::1
      2001:DB8:1:2::1
Serial0/0/0 [up/up] FE80::1
      2001:DB8:1:A001::2
Vlan1 [administratively down/down]
      unassigned
Router#show ipv6 route
IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
      I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS
      summary
      O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF
      ext 2
      ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
      D - EIGRP, EX - EIGRP external
C 2001:DB8:1:1::/64 [0/0]
      via ::, FastEthernet0/0
L 2001:DB8:1:1::1/128 [0/0]
      via ::, FastEthernet0/0
C 2001:DB8:1:2::/64 [0/0]
      via ::, FastEthernet0/1
L 2001:DB8:1:2::1/128 [0/0]
      via ::, FastEthernet0/1
C 2001:DB8:1:A001::/64 [0/0]
      via ::, Serial0/0/0
L 2001:DB8:1:A001::2/128 [0/0]
```

```
      via ::, Serial0/0/0
L FF00::/8 [0/0]
via ::, Null0
Router#
```

Pada ISP

```
Router>enable
Router#show ipv6 interface brief
FastEthernet0/0 [administratively down/down]
      unassigned
FastEthernet0/1 [administratively down/down]
      unassigned
Serial0/0/0 [up/up]
      FE80::1
      2001:DB8:1:A001::1
Vlan1 [administratively down/down]
      unassigned
Router#show ipv6 route
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S 2001:DB8:1:1::/64 [1/0]
      via FE80::1, Serial0/0/0
S 2001:DB8:1:2::/64 [1/0]
      via FE80::1, Serial0/0/0
C 2001:DB8:1:A001::/64 [0/0]
      via ::, Serial0/0/0
L 2001:DB8:1:A001::1/128 [0/0]
      via ::, Serial0/0/0
L FF00::/8 [0/0]
      via ::, Null0
Router#
```

Router 1:

```
R1#sh ipv6 int br
FastEthernet0/0            [up/up]
    FE80::1
    2001:DB8:1:1::1
FastEthernet0/1            [up/up]
    FE80::1
    2001:DB8:1:2::1
Serial0/0/0                [up/up]
    FE80::1
    2001:DB8:1:A001::2
Serial0/0/1                [administratively down/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
R1#sh ipv6 ro
IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:DB8:1:1::/64 [0/0]
     via FastEthernet0/0, directly connected
L   2001:DB8:1:1::1/128 [0/0]
     via FastEthernet0/0, receive
C   2001:DB8:1:2::/64 [0/0]
     via FastEthernet0/1, directly connected
L   2001:DB8:1:2::1/128 [0/0]
     via FastEthernet0/1, receive
C   2001:DB8:1:A001::/64 [0/0]
     via Serial0/0/0, directly connected
L   2001:DB8:1:A001::2/128 [0/0]
     via Serial0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

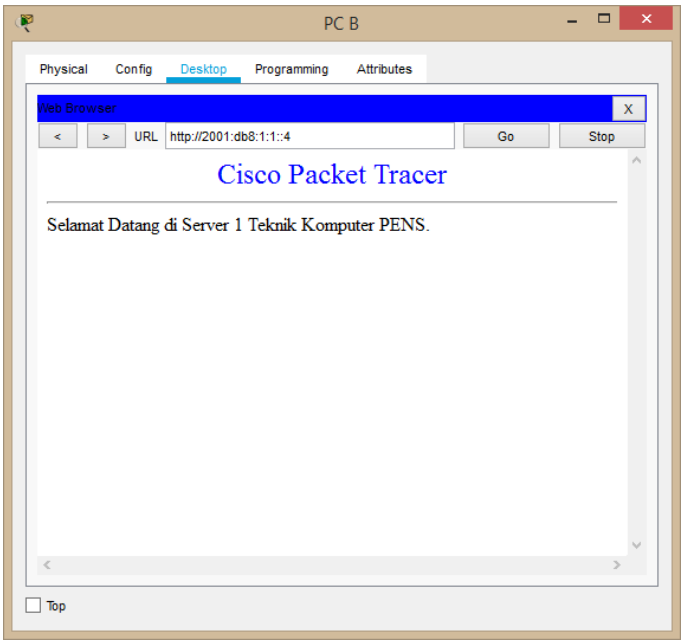
Router ISP:

```
ISP(config)#do sh ipv6 int br
FastEthernet0/0            [administratively down/down]
    unassigned
FastEthernet0/1            [administratively down/down]
    unassigned
Serial0/0/0                [up/up]
    FE80::1
    2001:DB8:1:A001::1
Serial0/0/1                [administratively down/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
ISP(config)#do sh ipv6 ro
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   2001:DB8:1:1::/64 [1/0]
     via FE80::1, Serial0/0/0
S   2001:DB8:1:2::/64 [1/0]
     via FE80::1, Serial0/0/0
C   2001:DB8:1:A001::/64 [0/0]
     via Serial0/0/0, directly connected
L   2001:DB8:1:A001::1/128 [0/0]
     via Serial0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

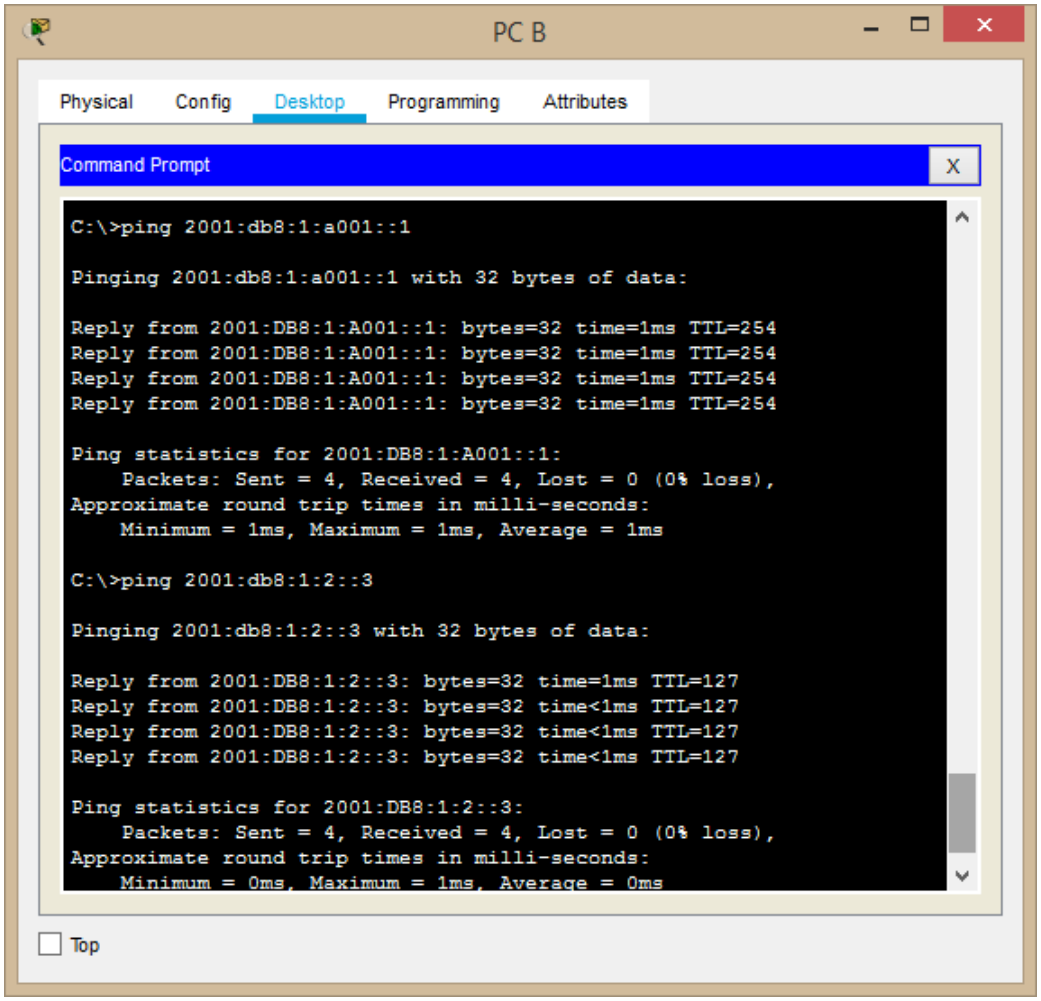
6. Lakukan Pengujian Konektivitas dari *Client*

   Buka *web browser* dari PC B kemudian ketikkan alamat IPv6 dari Server 1, kemudian klik Go.

   Lakukan hal yang sama unuk menguji konektivitas dari PC *client* lain semua server.

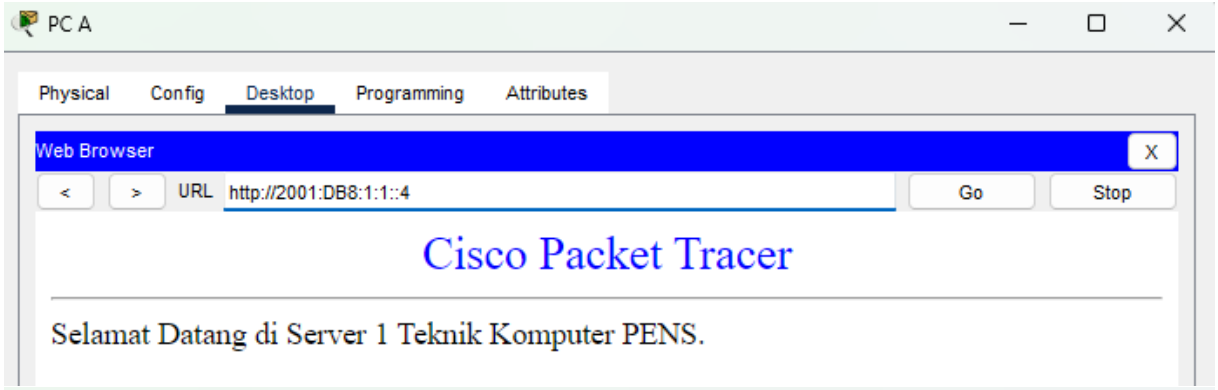


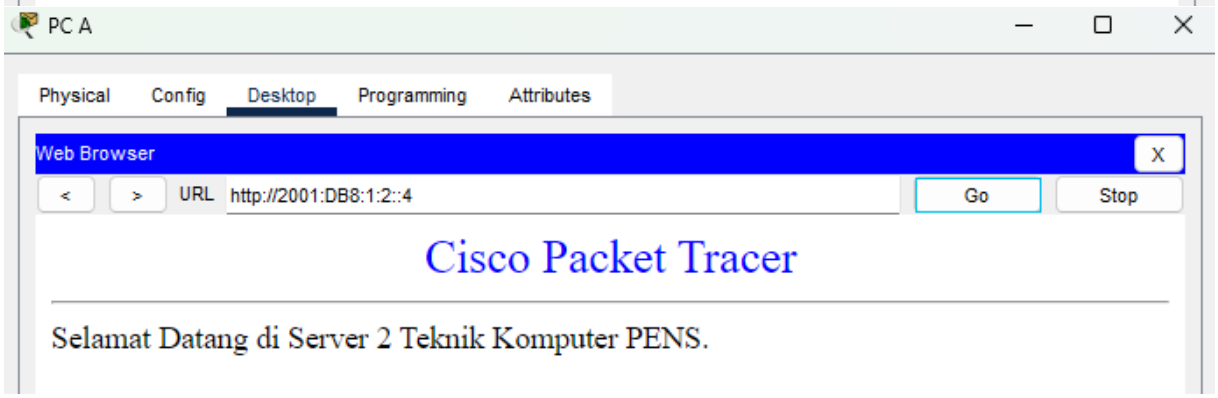
Selanjutnya lakukan pengujian tes *ping* antar *client* dank e ISP. Lakukan untuk semua *client.*



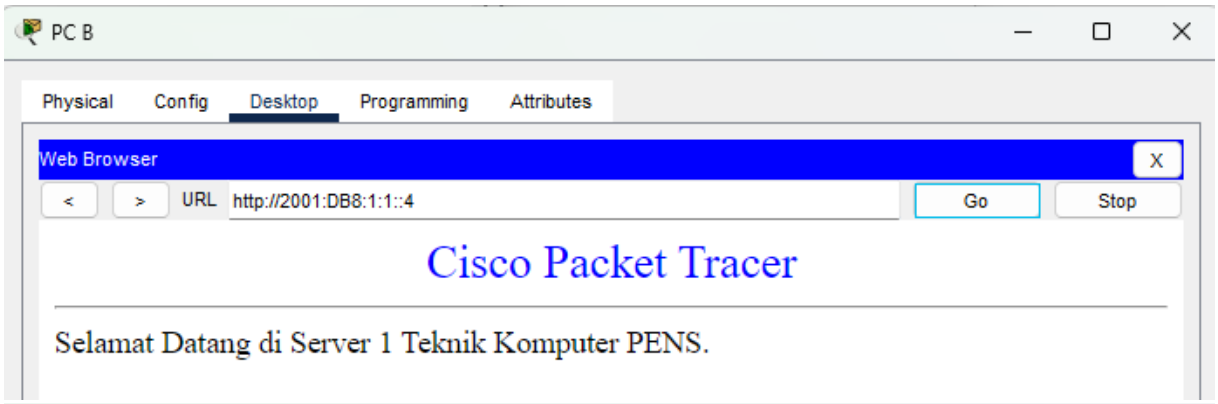
Pengujian web browser

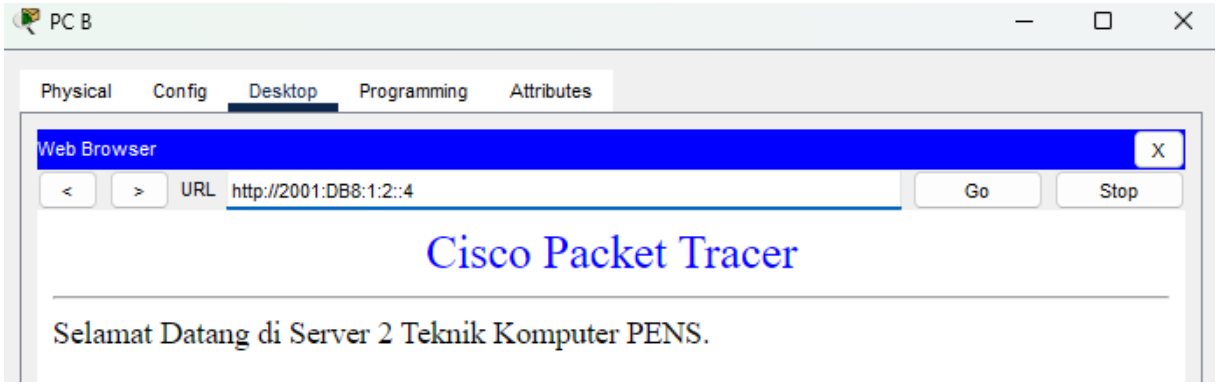
PC A



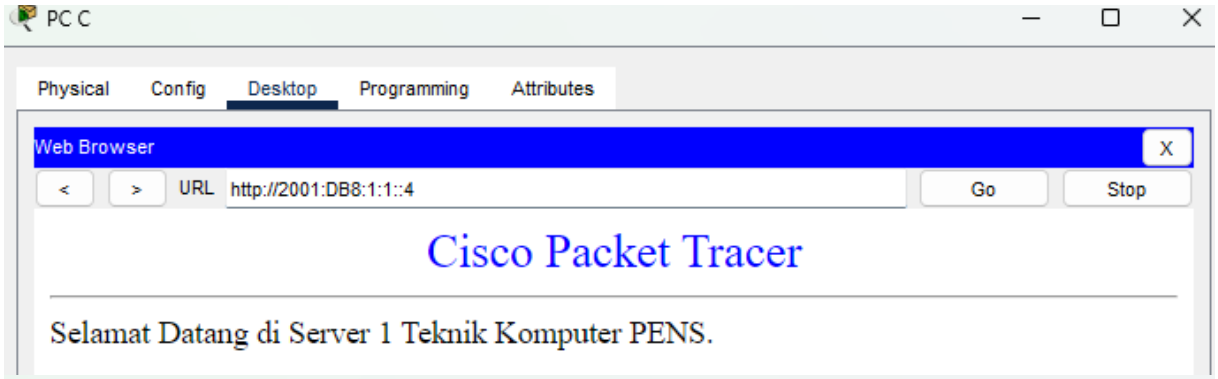


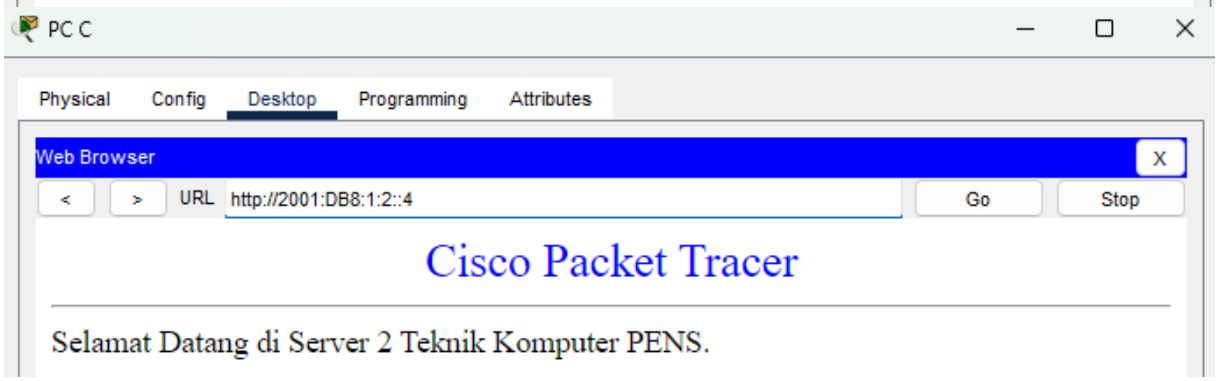
PC B



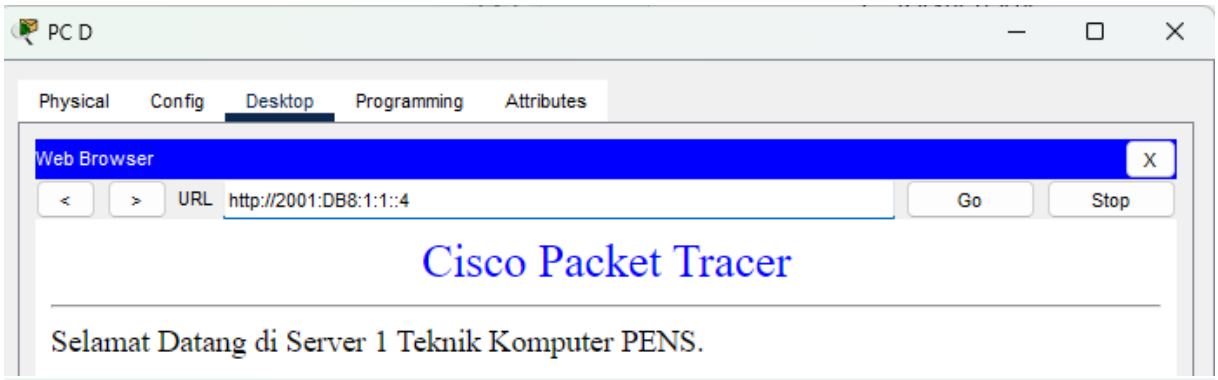


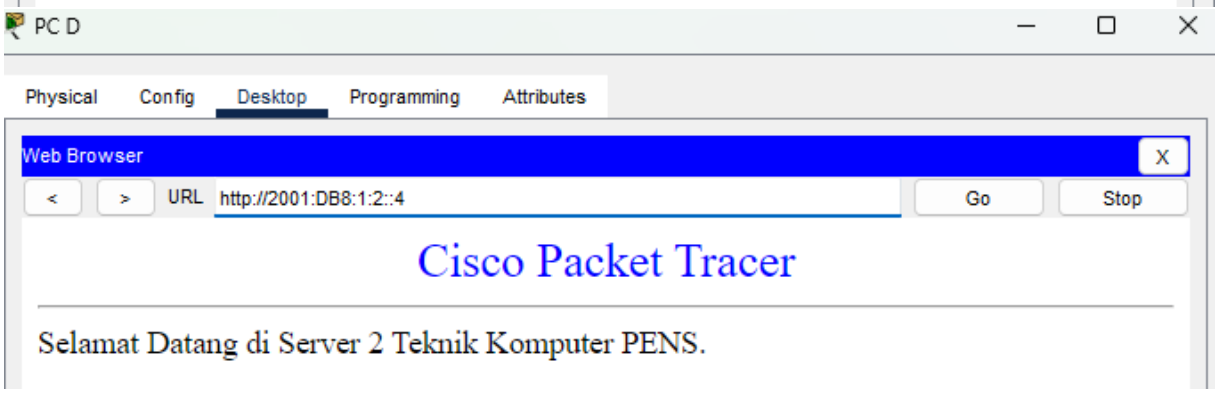
PC C


PC C
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://2001:DB8:1:1::4
Go
Stop
Cisco Packet Tracer
Selamat Datang di Server 1 Teknik Komputer PENS.


PC C
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://2001:DB8:1:2::4
Go
Stop
Cisco Packet Tracer
Selamat Datang di Server 2 Teknik Komputer PENS.

PC D


PC D
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://2001:DB8:1:1::4
Go
Stop
Cisco Packet Tracer
Selamat Datang di Server 1 Teknik Komputer PENS.


PC D
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://2001:DB8:1:2::4
Go
Stop
Cisco Packet Tracer
Selamat Datang di Server 2 Teknik Komputer PENS.

Pengujian ping
PC B ke PC lain serta ke Router ISP

PC B

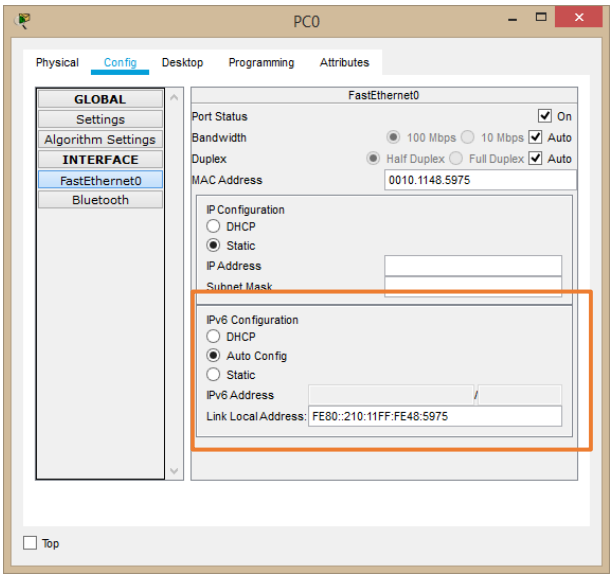
Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 2001:DB8:1:1::3

Pinging 2001:DB8:1:1::3 with 32 bytes of data:

Reply from 2001:DB8:1:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:1::3: bytes=32 time=1ms TTL=128

Ping statistics for 2001:DB8:1:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:DB8:1:2::2

Pinging 2001:DB8:1:2::2 with 32 bytes of data:

Reply from 2001:DB8:1:2::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::2: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:1:2::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:DB8:1:2::3

Pinging 2001:DB8:1:2::3 with 32 bytes of data:

Reply from 2001:DB8:1:2::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::3: bytes=32 time=1ms TTL=127

Ping statistics for 2001:DB8:1:2::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:DB8:1:A001::1

Pinging 2001:DB8:1:A001::1 with 32 bytes of data:

Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=64ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:DB8:1:A001::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 64ms, Average = 16ms
```

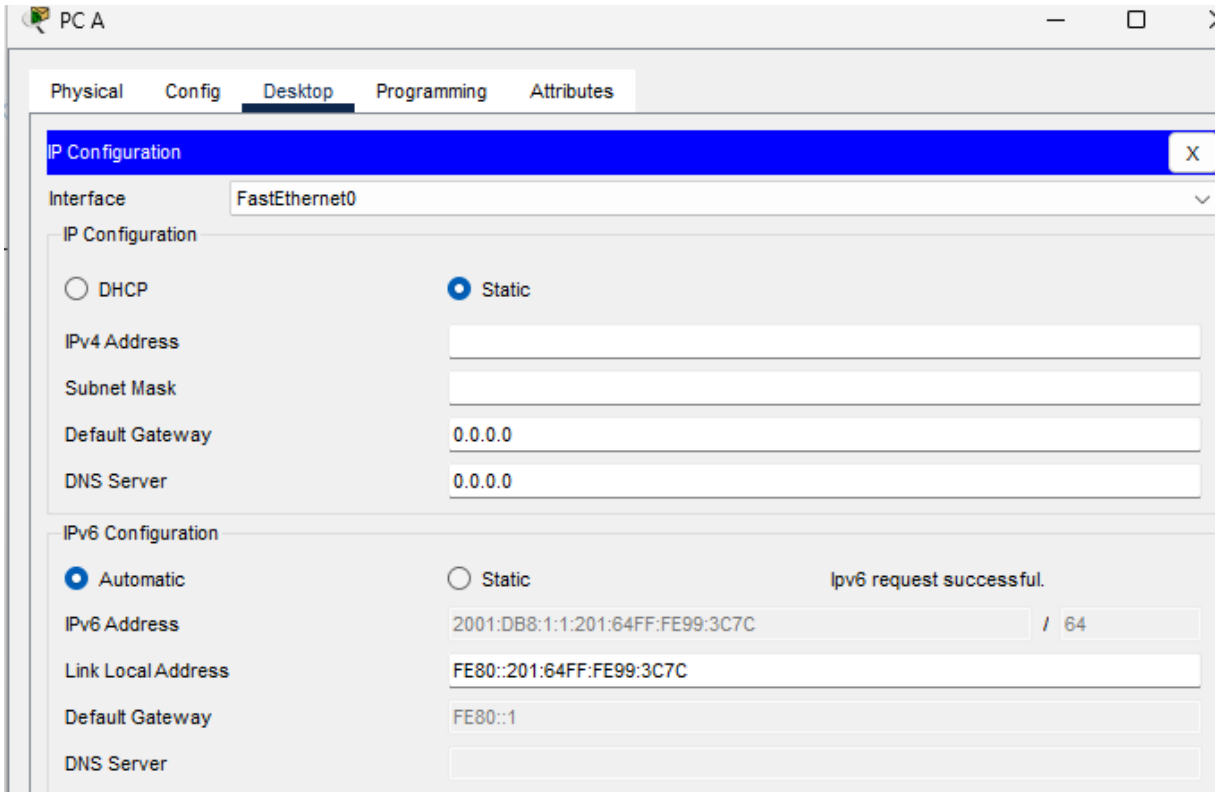
V. Tugas

Lakukan percobaan untuk topologi yang sama dengan konfigurasi IPv6 secara *Auto-config*

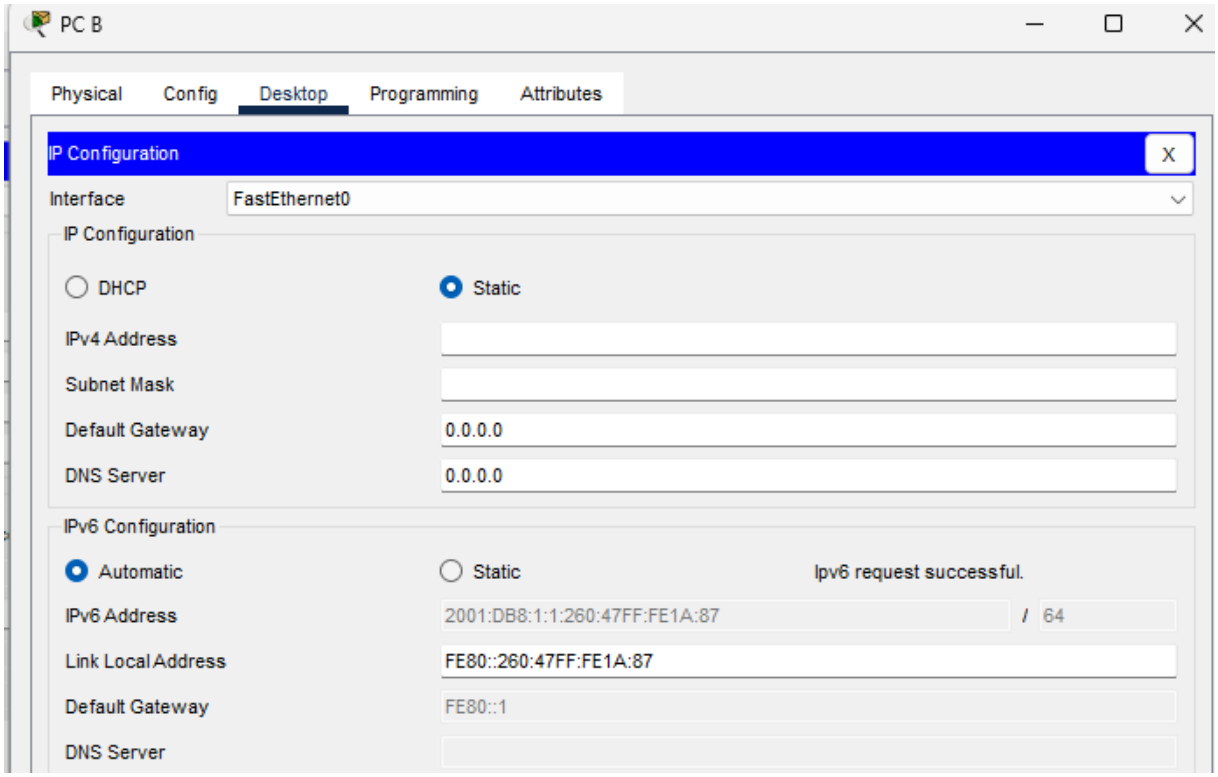


Addressing pada PC dengan automatic:

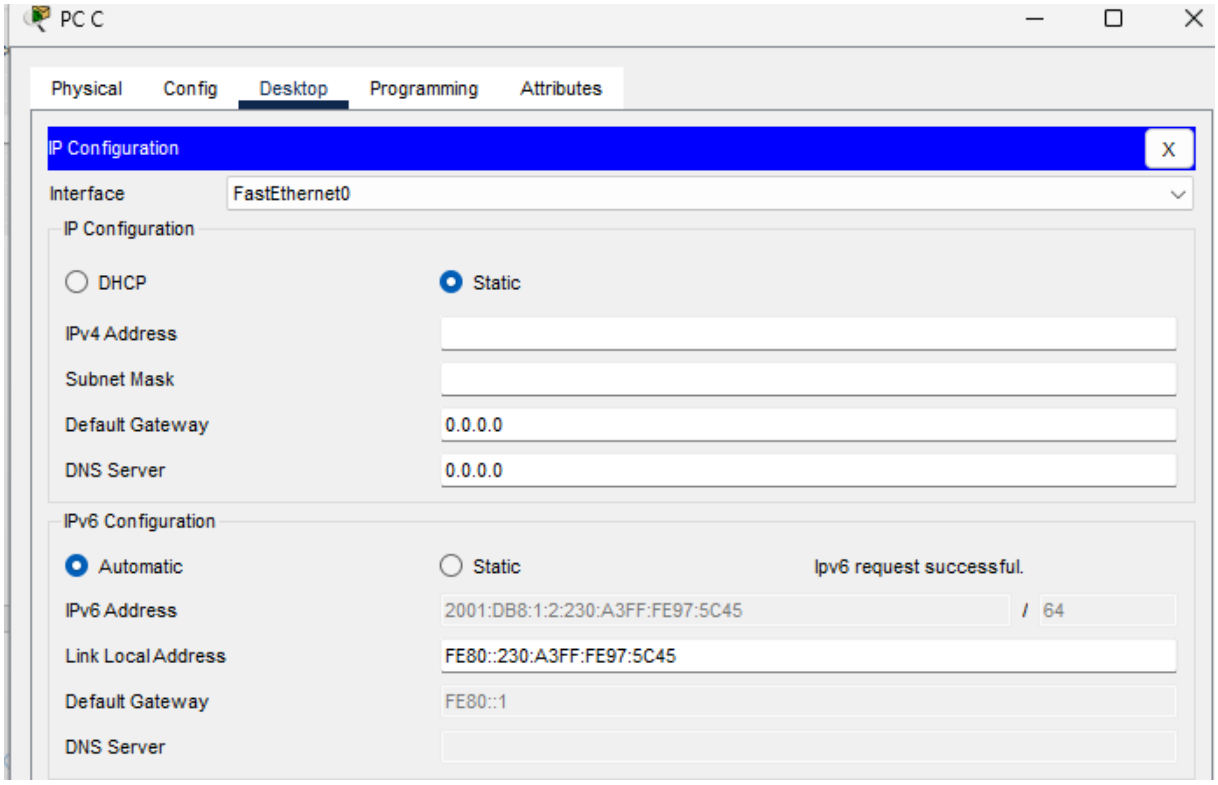
PC A



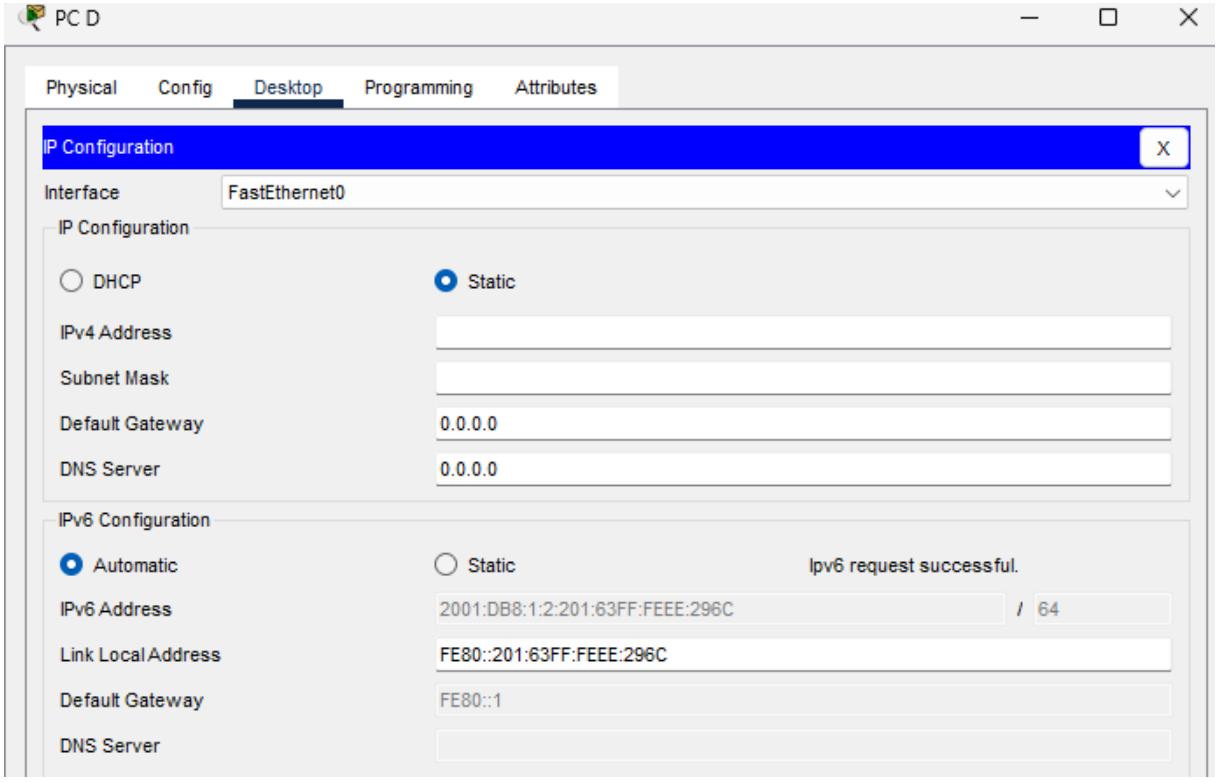
PC B


PC B
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
Ipv6 request successful.
IPv6 Address
2001:DB8:1:1:260:47FF:FE1A:87
/ 64
Link Local Address
FE80::260:47FF:FE1A:87
Default Gateway
FE80::1
DNS Server

PC C

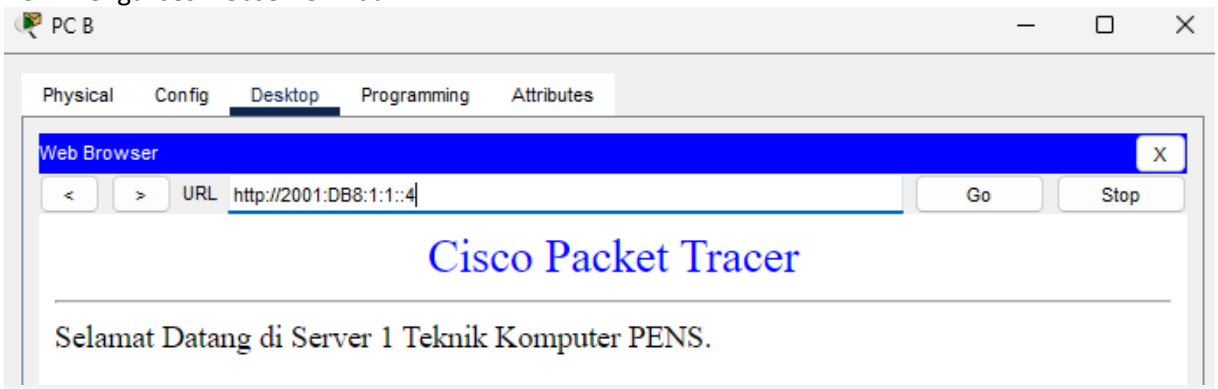

PC C
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
Ipv6 request successful.
IPv6 Address
2001:DB8:1:2:230:A3FF:FE97:5C45
/ 64
Link Local Address
FE80::230:A3FF:FE97:5C45
Default Gateway
FE80::1
DNS Server

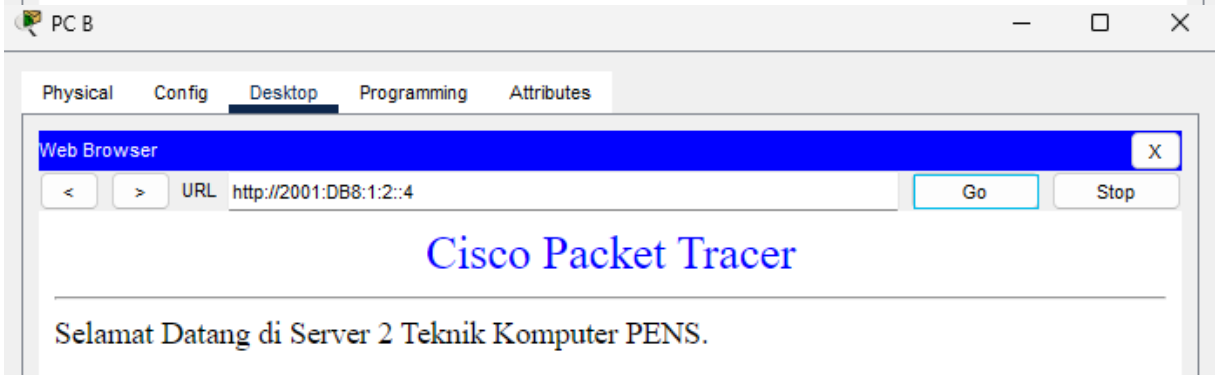
PC D



Pengujian salah satu PC pada kedua webserver dan ping menuju end devices lain:

PC B mengakses webserver 1 dan 2





PC B mengakses PC lain

PC B

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1:1:201:64FF:FE99:3C7C

Pinging 2001:DB8:1:1:201:64FF:FE99:3C7C with 32 bytes of data:

Reply from 2001:DB8:1:1:201:64FF:FE99:3C7C: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:1:201:64FF:FE99:3C7C: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1:1:201:64FF:FE99:3C7C: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:1:201:64FF:FE99:3C7C: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1:1:201:64FF:FE99:3C7C:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:DB8:1:2:230:A3FF:FE97:5C45

Pinging 2001:DB8:1:2:230:A3FF:FE97:5C45 with 32 bytes of data:

Reply from 2001:DB8:1:2:230:A3FF:FE97:5C45: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2:230:A3FF:FE97:5C45: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2:230:A3FF:FE97:5C45: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2:230:A3FF:FE97:5C45: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:1:2:230:A3FF:FE97:5C45:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:DB8:1:2:201:63FF:FEEE:296C

Pinging 2001:DB8:1:2:201:63FF:FEEE:296C with 32 bytes of data:

Reply from 2001:DB8:1:2:201:63FF:FEEE:296C: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2:201:63FF:FEEE:296C: bytes=32 time=1ms TTL=127
Reply from 2001:DB8:1:2:201:63FF:FEEE:296C: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2:201:63FF:FEEE:296C: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:1:2:201:63FF:FEEE:296C:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Scenario lain dengan penggunaan EUI-64 untuk konfigurasi IPv6 di router R1, khususnya di interface FastEthernet yang mengarah ke subnet berbeda.

Konfigurasi pada R1

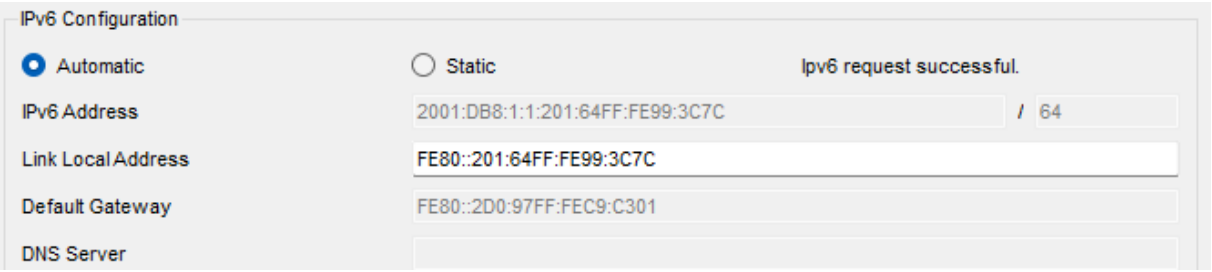
```
R1(config-if)#int f0/0
R1(config-if)#no ipv6 add
R1(config-if)#ipv6 address 2001:DB8:1:1::/64 eui-64
R1(config-if)#int f0/1
R1(config-if)#no ipv6 add
R1(config-if)#ipv6 address 2001:DB8:1:2::/64 eui-64
```

Verifikasi dengan "`show ipv6 interface brief`"

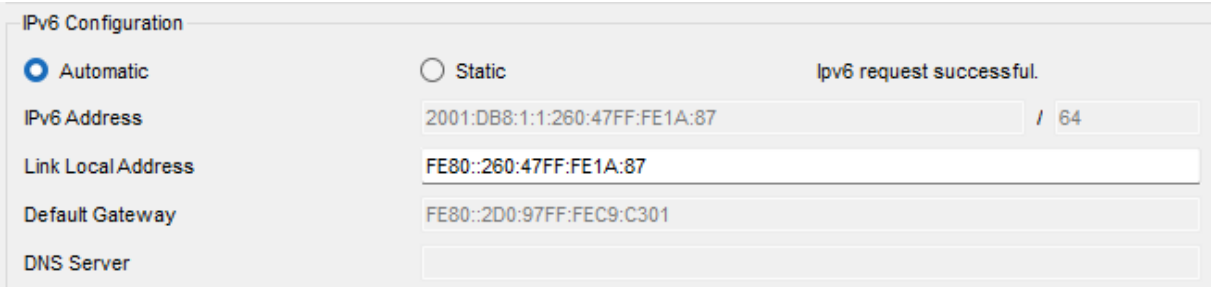
```
R1(config)#do sh ipv6 int br
FastEthernet0/0            [up/up]
    FE80::2D0:97FF:FEC9:C301
    2001:DB8:1:1:2D0:97FF:FEC9:C301
FastEthernet0/1            [up/up]
    FE80::2D0:97FF:FEC9:C302
    2001:DB8:1:2:2D0:97FF:FEC9:C302
Serial0/0/0                [up/up]
    FE80::1
    2001:DB8:1:A001::2
```

Addressing ulang pada PC dengan automatic

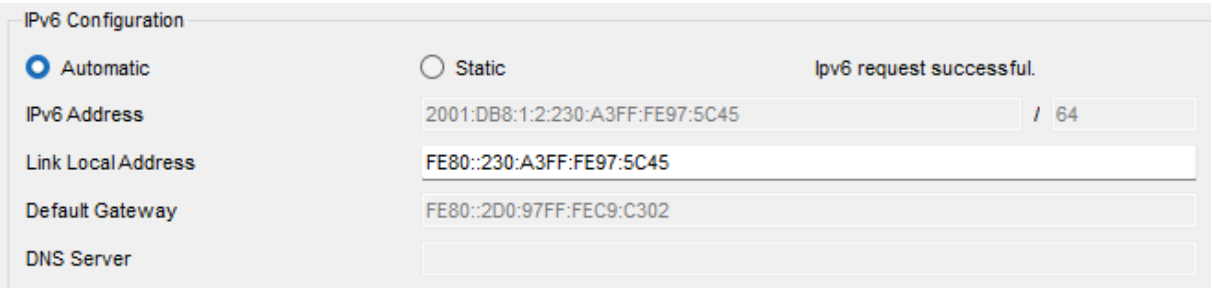
PC A



PC B



PC C

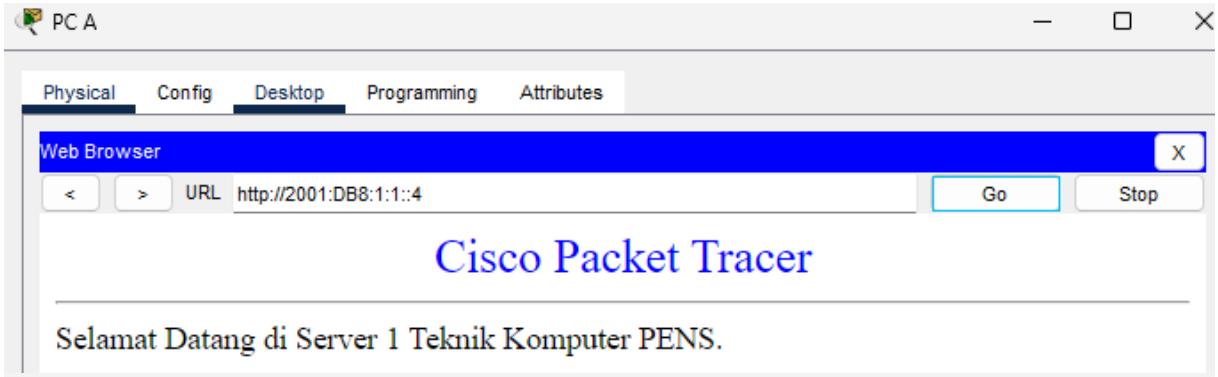


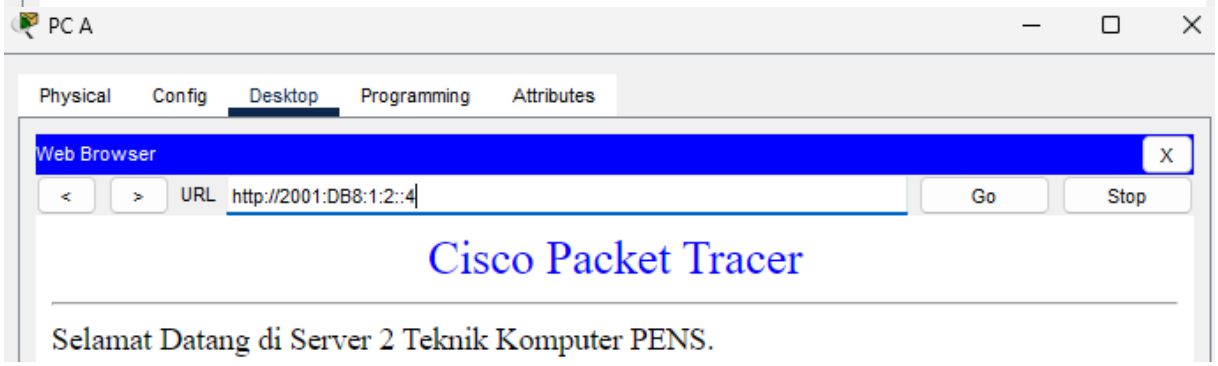
PC D

Verifikasi dari PC A via simple PDU

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC A	PC B	ICM...		0.000	N	0	(edit)
	Successful	PC A	PC C	ICM...		0.000	N	1	(edit)
	Successful	PC A	PC D	ICM...		0.000	N	2	(edit)

Verifikasi webserver





VI. Analisa

Pada praktikum dengan konfigurasi pengalamatan Ipv6, dapat terlihat bahwa format alamat IPv6 sudah berbeda jauh dari IPv4, memiliki panjang 128 bit yang dibagi menjadi delapan grup hexadecimal, sehingga IPv6 dapat mendukung jumlah alamat yang sangat banyak. Pengalamatan diawal pada seluruh end devices dan router dilakukan secara manual. Server 1, PC A, dan PC B berada pada subnet 2001:DB8:1:1::/64, sedangkan PC C, PC D, dan server 2 berada pada subnet 2001:DB8:1:2::/64, semua menggunakan default gateway link-local FE80::1. Router R1 memiliki tiga interface utama, f0/0 dengan 2001:DB8:1:1::1/64, f0/1 dengan 2001:DB8:1:2::1/64, dan s0/0/0 dengan 2001:DB8:1:A001::2/64. Router ISP terhubung dengan R1 via s0/0/0 dengan alamat 2001:DB8:1:A001::1/64. Penggunaan prefix /64 di setiap link bertujuan untuk memastikan ruang alamat yang memadai untuk SLAAC atau manual host ID.

Konfigurasi pada router dimulai dengan perintah *ipv6 unicast-routing*, yang mengaktifkan kemampuan router untuk meneruskan paket IPv6. Masuk ke setiap interface, contohnya interface f0/0, lalu *ipv6 enable* memunculkan alamat link-local otomatis (FE80::/64) selain yang ditetapkan manual dengan *ipv6 address FE80::1 link-local*. Alamat link-local ini penting karena digunakan sebagai default gateway bagi end devices dan untuk Neighbor Discovery (ND). Perintah *ipv6 address 2001:DB8:1:1::1/64* menetapkan global unicast address sesuai subnet, dan no shutdown mengaktifkan interface. Langkah tersebut juga dilakukan pada f0/1 dan s0/0/0.

```
R1(config)#ipv6 unicast-r
R1(config)#int f0/0
R1(config-if)#ipv6 en
R1(config-if)#ipv6 add 2001:DB8:1:1::1/64
R1(config-if)#ipv6 add FE80::1 link-l
R1(config-if)#no sh
R1(config-if)#int f0/1
R1(config-if)#ipv6 en
R1(config-if)#ipv6 add 2001:DB8:1:2::1/64
R1(config-if)#ipv6 add FE80::1 link-l
R1(config-if)#no sh
R1(config-if)#int s0/0/0
R1(config-if)#ipv6 en
R1(config-if)#ipv6 add 2001:DB8:1:A001::2/64
R1(config-if)#ipv6 add FE80::1 link-l
R1(config-if)#no sh
```

Kemudian di router ISP dibuat static routes dengan *ipv6 route 2001:DB8:1:1::/64 s0/0/0 FE80::1* dan *ipv6 route 2001:DB8:1:2::/64 s0/0/0 FE80::1* agar paket dari kedua subnet dikirim melalui link-local router R1.

```
ISP(config)#ipv6 unicast-r
ISP(config)#int s0/0/0
ISP(config-if)#ipv6 en
ISP(config-if)#ipv6 add 2001:DB8:1:A001::1/64
ISP(config-if)#ipv6 add fe80::1 link-l
ISP(config-if)#no sh
ISP(config-if)#ipv6 unicast-r
ISP(config)#ipv6 route 2001:db8:1:1::/64 s0/0/0 fe80::1
ISP(config)#ipv6 route 2001:db8:1:2::/64 s0/0/0 fe80::1
```

Seluruh perangkat diuji konektivitasnya menggunakan perintah ping dan traceroute. Hasil pengujian menunjukkan bahwa PC A–D dan kedua server berhasil saling berkomunikasi meski berada di subnet berbeda, serta mampu mengakses layanan web pada masing-masing server. Selanjutnya, end devices diubah ke mode SLAAC ("automatic"), sehingga alamat IPv6 dihasilkan secara otomatis dengan menggabungkan prefix yang diumumkan Router R1 dan host ID berbasis MAC. Sebagai contoh, PC A

memperoleh alamat 2001:DB8:1:1:0201:64FF:FE99:3C7C dengan gateway FE80::1. Meskipun metode alokasi alamat berbeda, hasil uji konektivitas tetap konsisten dengan konfigurasi manual, yang menunjukkan keberhasilan mekanisme SLAAC dalam menyediakan end-to-end connectivity tanpa konfigurasi manual.

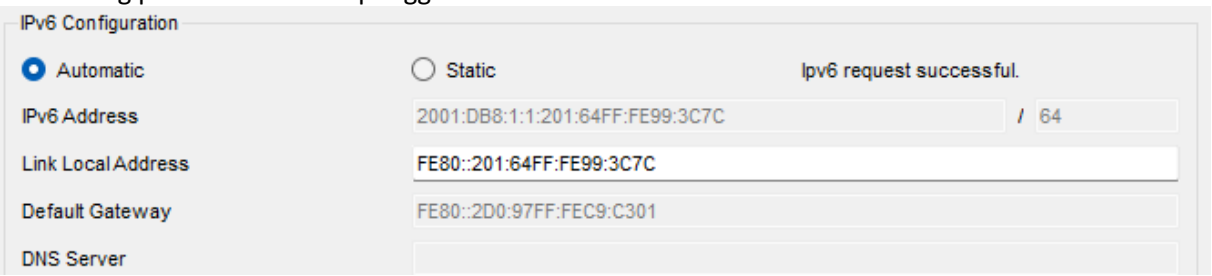
Sebagai skenario lanjutan, Router R1 dikonfigurasi ulang menggunakan EUI-64 pada interface FastEthernet untuk menghasilkan host ID secara otomatis:

```
R1(config-if)#int f0/0
R1(config-if)#no ipv6 add
R1(config-if)#ipv6 address 2001:DB8:1:1::/64 eui-64
R1(config-if)#int f0/1
R1(config-if)#no ipv6 add
R1(config-if)#ipv6 address 2001:DB8:1:2::/64 eui-64
```

Setelah konfigurasi ini, end devices masih memperoleh alamat global yang sama lewat SLAAC, namun default gateway link-local berubah menjadi nilai EUI-64, contohnya FE80::2D0:97FF:FEC9:C301 pada subnet kiri dan FE80::2D0:97FF:FEC9:C302 pada subnet kanan.

Addressing pada PC A sebelum penggunaan EUI-64

Addressing pada PC A setelah penggunaan EUI-64



Server yang tetap menggunakan alamat statis dengan gateway FE80::1 juga terus dapat berkomunikasi karena alamat link-local tersebut masih aktif di router. Bagaimana jika server menggunakan automatic juga? Apabila dialihkan ke mode SLAAC, server akan memilih link-local hasil EUI-64, dan jaringan tetap beroperasi normal.

VII. Kesimpulan

Pengalamatan IPv6 memungkinkan penerapan skema alamat yang lebih fleksibel dan skalabel dengan memisahkan prefix /64 dan host ID, sehingga perangkat dapat dikonfigurasi secara manual maupun otomatis tanpa mengorbankan konektivitas. Dalam praktikum ini, baik metode manual, automatic (SLAAC), maupun EUI-64 pada router R1 terbukti menghasilkan alamat konsisten dan memastikan link-local sebagai default gateway terkelola dengan baik. Hasil pengujian menunjukkan semua end devices dan server tetap saling terhubung dan dapat mengakses layanan web di subnet berbeda. Dengan demikian, IPv6 memberikan kemudahan operasional sekaligus menjaga kestabilan jaringan tanpa perlu intervensi manual berulang.